

ATOKA, OK, USA

WMO: 722051

Lat: 34.398N

Lon: 96.148W

Elev: 180

StdP: 99.18

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53932

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.6	-5.2	-13.0	1.2	-3.1	-11.3	1.5	-0.6	9.3	14.1	8.7	13.0	1.6	10	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	38.7	23.6	37.0	24.0	35.1	24.1	27.5	31.7	26.4	31.0	25.5	31.3	2.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.3	22.2	30.3	25.2	20.7	28.8	23.9	19.2	28.0	88.5	32.2	83.2	31.6	79.3	31.3	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.3	7.4	6.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-11.9	40.0	2.5	3.4	-13.7	42.4	-15.2	44.4	-16.6	46.3	-18.4	48.8		
			-12.4	27.2	2.3	1.2	-14.0	28.1	-15.4	28.8	-16.7	29.4	-18.3	30.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	5.5	8.0	13.2	16.9	21.2	26.5	28.3	27.8	24.0	17.7	11.4	6.3
	DBStd	9.21	5.42	6.07	5.26	4.53	3.85	2.62	2.65	3.15	3.62	5.02	5.39	4.89
	HDD10.0	481	161	100	27	6	0	0	0	0	0	4	48	136
	HDD18.3	1619	399	291	171	77	18	0	0	0	3	74	214	373
	CDD10.0	3141	20	46	128	213	347	494	566	553	419	244	89	21
	CDD18.3	1240	0	3	13	35	107	244	308	295	172	56	7	1
	CDH23.3	13669	4	17	101	317	967	2653	3771	3466	1770	559	42	3
	CDH26.7	6744	0	2	13	64	323	1288	2114	1911	828	198	3	0
(14) Wind	WSAvg	2.2	2.4	2.8	2.9	2.8	2.3	2.0	1.6	1.4	1.5	1.9	2.3	2.1
(15) Precipitation	PrecAvg	1197	61	59	105	127	148	123	79	74	125	116	72	79
	PrecMax	1685	201	137	210	272	277	237	230	188	242	323	217	208
	PrecMin	749	0	0	21	19	16	41	0	25	33	24	13	3
	PrecStd	236	49	41	51	55	81	60	63	39	60	77	57	55
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	25.3	27.9	31.4	33.6	37.5	41.1	42.5	37.5	32.6	27.1	22.8
		MCWB	13.6	16.6	18.7	21.0	23.5	24.2	23.7	23.1	22.2	21.5	17.9	15.9
	2%	DB	21.0	22.7	26.2	27.9	32.3	36.0	38.9	39.0	35.0	31.2	24.7	20.1
		MCWB	13.0	15.7	17.6	19.2	23.1	24.2	23.8	23.5	23.0	21.5	16.4	14.4
	5%	DB	17.8	20.9	23.8	27.0	30.2	34.0	37.0	37.1	32.9	28.9	22.5	17.4
		MCWB	12.6	15.1	16.7	18.7	22.3	23.4	24.2	24.0	22.4	20.1	16.3	12.4
	10%	DB	15.2	18.0	22.3	25.1	27.9	32.6	35.1	34.8	31.4	27.0	20.9	14.2
		MCWB	9.4	13.6	16.1	18.2	21.1	24.2	24.5	24.1	22.1	19.0	16.0	9.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.9	19.0	20.6	22.8	25.7	27.9	28.8	28.9	26.3	24.2	20.3	18.6
		MCDB	19.6	23.1	25.3	27.8	31.9	32.1	34.6	34.1	30.9	30.9	23.2	21.6
	2%	WB	15.7	17.7	19.2	21.6	24.3	27.0	27.4	27.5	24.8	22.7	19.1	16.2
		MCDB	18.6	20.7	24.1	26.3	29.6	31.0	32.0	31.8	30.3	28.4	22.6	18.4
	5%	WB	12.9	16.5	18.3	20.6	23.3	25.8	26.1	26.1	23.8	21.8	17.9	13.5
		MCDB	16.9	19.3	22.4	24.8	28.3	30.9	31.4	31.1	29.8	26.6	20.7	15.9
	10%	WB	10.3	14.3	17.3	19.5	22.4	25.1	25.2	25.3	23.1	20.6	16.4	11.2
		MCDB	14.6	18.2	20.6	23.7	26.9	30.9	31.7	31.4	29.1	24.1	19.6	13.9
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	12.1	12.2	12.4	11.1	11.1	12.1	11.7	12.7	12.8	12.8	11.1
		MCDBR	17.1	15.0	14.1	13.7	12.8	12.8	14.5	14.5	14.2	14.6	13.7	15.2
	5% WB	MCWBR	10.7	9.3	7.6	7.0	5.7	4.3	3.8	3.7	4.7	6.4	8.1	9.9
		MCWBR	13.9	12.7	11.2	10.8	10.6	10.9	12.0	11.8	11.9	12.1	11.1	11.9
(40) Clear-Sky Solar Irradiance	taub	0.309	0.318	0.355	0.386	0.424	0.433	0.442	0.446	0.398	0.352	0.331	0.309	(40)
		2.491	2.484	2.405	2.328	2.286	2.298	2.313	2.283	2.371	2.487	2.486	2.499	(41)
	Ebn at Noon	903	932	918	900	865	854	844	834	862	881	868	879	(42)
		85	95	112	126	133	132	129	131	114	94	85	80	(43)
(44) All-Sky Solar Radiation	RadAvg	2.69	3.19	4.26	5.21	5.65	6.45	6.58	6.03	5.02	3.86	2.86	2.32	(44)
	RadStd	0.28	0.47	0.30	0.39	0.57	0.71	0.39	0.38	0.49	0.56	0.36	0.30	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+75

Nomenclature: See separate page